

GSoC 2026 Proposal

Advanced Benchmarking and Anatomically-Informed Segmentation for Coronary Artery Calcium (CAC)

Candidate: Shriyam Baloni
Organization: ML4Sci (PREDICT)
Project: Building and Comparing Segmentation Strategies for CAC
Mentors: Kathleen Butler, Anna Marie Love, Sergei Gleyzer, Harvey Hahn

1. Abstract

Coronary Artery Calcium (CAC) scoring is a critical independent predictor of Major Adverse Cardiovascular Events (MACE). However, automated quantification in non-contrast CT (NCCT) faces significant hurdles: severe class imbalance, high imaging noise, and frequent false positives from neighboring osseous structures. This project transitions CAC segmentation from a purely data-driven task to an anatomically-informed system. By leveraging an established registration framework—which achieves an **82.75% mean localization accuracy with an 86.6% pass rate**—I propose a pipeline that fuses high-performance 3D architectures with physiological constraints. This approach bridges data-driven learning with anatomical reasoning, enabling more reliable and clinically interpretable CAC quantification.

2. Problem Statement & Motivation

Accurate CAC scoring requires the isolation of sparse, high-intensity lesions within the coronary arteries. Current deep learning models often struggle with:

- a) **Anatomical Ambiguity:** Misidentifying rib or spinal density as CAC due to similar Hounsfield Unit (HU) profiles.
- b) **Clinical Inconsistency:** Purely pixel-wise losses (like Dice) do not account for the density-weighting required for Agatston scoring.
- c) **Lack of Attribution:** Standard segmentation fails to assign plaques to specific arteries (LAD, LCX, RCA), which is vital for clinical risk stratification.

This project addresses these gaps by **building a Vessel-Aware system that incorporates anatomical priors to constrain and refine predictions.**

3. Concrete Technical Approach

3.1 Preprocessing & Intensity Harmonization

The pipeline will utilize my established multi-stage preprocessing suite:

- **Metadata Bridge:** Stripping inconsistent physical metadata to canonical voxel space to **ensure stable initialization for the registration process.**
- **Intensity Harmonization:** HU clipping at $[-200, 600]$ followed by a linear rescale to $[0, 1]$. This preserves the air-tissue boundary and vessel contrast required for Mattes Mutual Information (MI) stability.

- **Mediastinal ROI Cropping:** Utilizing heart-specific masks to reduce search space by $\sim 70\%$ and eliminate peripheral skeletal interference.

3.2 Multi-Strategy Benchmarking Suite

I will implement and systematically compare three architectural classes:

1. **Efficiency Baseline:** 2.5D U-Net variants for high-throughput evaluation.
2. **SOTA Standard:** A fully configured **nnU-Net** (3D fullres) implementation.
3. **Global Contextual: Swin-UNETR** to leverage hierarchical Transformers for capturing long-range dependencies.

3.3 Innovation: Anatomically-Constrained Priors

Using the B-Spline registration centerlines developed in Phase 0, I will generate **Vessel Distance Fields (VDF)**, representing spatial proximity to coronary artery centerlines.

- **Consistency Loss:** A secondary loss term, $L_{Anatomical}$, will penalize predictions in anatomically implausible regions (e.g., outside coronary pathways).
- **Uncertainty Refinement:** Integration of **Monte Carlo (MC) Dropout** during inference to resolve ambiguous clusters using the VDF as a "tie-breaker."

4. Deliverables

- **Production-Ready CLI Tool:** A containerized module supporting NIfTI/DICOM inputs.
- **Automated Scoring Script:** An Agatston calculator using area and density multipliers.
- **Anatomical Attribution Module:** Plaque assignment to specific coronary branches.
- **System Scalability:** Designed for seamless integration into the PREDICT workflow and future dataset scalability.

5. Timeline with Milestones

Week	Phase	Milestone / Deliverable
Bonding	Community	Finalize data splits; CI/CD setup; integrate Phase 0 code.
W1–2	Baseline	Implement preprocessing suite and 2.5D efficiency baseline.
W3–4	SOTA	Train and evaluate nnU-Net on full COCA dataset.
W5–6	Transformers	Implement Swin-UNETR; conduct architecture ablation studies.
W7–8	Priors	Integrate VDFs; implement $L_{Anatomical}$ loss and MC Dropout.
W9–10	Clinical	Finalize Agatston scoring and Anatomical Attribution modules.
W11	Validation	Scaling tests; analyze FPR near bone; visual ROI overlay.
W12	Finalization	CLI packaging; finalize documentation and Development Report.

6. Evaluation Plan

- **Technical Metrics:** Dice Coefficient, Precision/Recall, and AUC.
- **Clinical Metrics:** Agreement between predicted and ground-truth Agatston scores.
- **Ablation Studies:** Evaluate contribution of ROI cropping, anatomical priors, and uncertainty refinement.
- **Visual Validation:** Overlay calcium segmentation with heart ROI to ensure anatomical consistency (as recommended by mentors).

7. Preparation, Prior Work, & Mentor Guidance

I have successfully completed the following "Phase 0" tasks:

- **Segmentation (Task 1):** Attention-Gated 3D U-Net achieving a **Dice score of 0.8513**.
GitHub Repository (Tasks 1): [Codebase with Task 1 \(Segmentation\)](#)
- **Registration (Task 3):** Mean calcium localization accuracy of **82.75%** and an **86.6% pass rate** (peak: 99.98%).
GitHub Repository (Tasks 3): [Task 3 \(Registration\) implementations, results, and analysis](#)

[Codebase with all Task's implementation](#)

8. Risks, Fallbacks, and Execution Guarantee

- **Risk:** Registration noise in low-contrast scans. *Fallback:* Use established label-centroid translation pre-alignment (Stage 0).
- **Execution Guarantee:** The core deliverable—a validated, anatomy-aware segmentation pipeline—will be completed by Week 10.

Acknowledgments

I would like to thank **Kathleen (Katy) Butler** for her guidance during the development of this proposal. Her feedback on prioritizing usability, validation, and scalability helped shape the direction of this work. In particular, her suggestions on improving evaluation and incorporating meaningful validation strategies were valuable in refining both the technical approach and its clinical relevance.